TELIPORT
Script Your Future
Season 4

TATA
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CASE STUDY 1 / CONNECTED FLEET PREDICTIVE MAINTENANCE

FlairGuard AI — A Verified Connected-Fleet Reliability Platform

Causal RUL prediction, telemetry-fault recovery and auditable maintenance decisions under constrained service capacity.

Team: FlairGuard | Case Study: 1

NASA C-MAPSS → REGIME-AWARE PROGNOSTICS → TELEMETRY QUALITY & RECOVERY → DETERMINISTIC MAINTENANCE DECISIONS → PERSISTENT WORK ORDERS & AUDIT

NASA C-MAPSS	→	Regime Prognostics	→	Quality & Recovery	→	Deterministic Decisions	→	Work Orders & Audit
21 sensors, 3 settings Past-only rolling windows		LightGBM / HGB / Ridge Regime-aware normalization		Missingness & staleness Abstention & 6 recovery methods		RUL & quality thresholds Human review for exceptions		Priority, deduplication SQLite audit & recovery

FD001/FD002 operational serving; FD003/FD004 retrospective research coverage · MATLAB toolbox comparison integrated

TRUST

Engine-disjoint training and causal inference; decision-horizon risk evaluation; six telemetry-recovery methods tested across eight controlled conditions.

VALUE

FD002 trigger-outage scenario: 0/80 modeled failures with regime imputation versus 14/80 with bounded countdown.

Scenario-specific retrospective result; no universal superiority claim.

EXECUTION

Working FastAPI + React operator platform, authenticated work-order workflows, restart-safe replay, model-version pinning, real MLflow training and rollback verification.

Real Chromium browser workflow and Docker restart checks completed.

114 passing tests · 332 primary policy runs · 17,822 scored engine episodes from 169 distinct source engines · Independent event-ledger verification · MATLAB toolbox comparison integrated

NASA C-MAPSS retrospective simulation; costs are hypothetical. Scientific model promotion requires fresh acceptance data.